

AI (Artificial Intelligence):

AI is the simulation of human intelligence in machines, allowing them to think, learn, and make decisions.

AI enables computers to analyze data, recognize patterns, and solve problems without explicit human instructions.

Characteristics of AI:

1. **Learning** – Improved performance based on experiences
2. **Reasoning** – Logical decision-making
3. **Problem-Solving** – Finding efficient solutions
4. **Perception** – AI processes and interprets visual, audio, and sensor data

AI Problems:

1. Game Playing and Theorem Proving

- Game playing and theorem proving are done by using AI.
- Computers can perform well at these tasks by being fast at exploring a large number of solution paths and then selecting the best one.
 - *Examples:* Chess, Tic-Tac-Toe

2. Common Sense Reasoning

- Common sense reasoning means understanding everyday situations like humans.
 - *Example:*
 - If you drop a glass, it will break.
 - If it's raining, people carry umbrellas.

3. Perception (Vision & Speech)

- AI sees and understands images (colored or uncolored), language too, and processes speech (speech recognition).
- **Ex:**
 - Vision → Face recognition, object detection
 - Speech → Alexa, Siri, Google Assistant

5. Natural Language Understanding

- NLU is a subfield of Artificial Intelligence.
- It enables machines to understand and interpret **human language** as it is spoken or written.
- It involves the ability to:
 - Comprehend meaning,
 - Recognize context,
 - Understand grammar and syntax,
 - Identify intent,
 - Handle ambiguity in language.

AI Techniques

1. Knowledge Representation

- Used to **capture knowledge** about the **domain/world**.
- Helps AI systems to **store, organize, and utilize** information to make intelligent decisions.

2. Search Algorithm

- Used in the **searching or finding of a solution** to a problem.
- AI uses search when there is **no direct or predefined solution**.
- It explores **all possible options** and selects the **optimal path** or decision.

3. Use of Knowledge

- AI solves **complex problems** using **prior knowledge**.
- It understands and applies **rules, facts, and logic** to make decisions.

Example:

- **Medical diagnosis AI** using patient history and symptoms to predict illness.

4. Abstraction

- AI focuses only on the **important details**, ignoring unessential ones.
- This **simplifies complex problems** and increases **efficiency**.

Example:

- In **image recognition**, AI focuses on **object shapes** and **ignores the background** to identify key features.

Foundations of AI

1. Philosophy & Logic

- AI involves understanding **knowledge** and how problems can be solved logically.
- Philosophy helps define **intelligence** and how machines can “think”.
- Logic helps an AI system make decisions based on **facts and rules**.

2. Mathematics & Statistics

- Mathematics provides a foundation for **algorithms and computations**.
- Statistics is crucial for AI to **analyze data** and make **predictions**.
- Key concepts include:
 - **Proofing theorems**
 - **Algorithms**
 - **Computation**
 - **Decidability**
 - **Reasoning from data**

3. Economics

- AI applies economic principles to decision-making.
- Concepts like maximizing **utility** (best possible outcome with minimal investment).
- AI can analyze available information to make **optimized decisions**.

4. Psychology

- Understands **human behavior, perception, and emotions**.

5. Computer Science & Engineering

- AI depends heavily on **programming algorithms** and system design.
- **Logic, interface theory**, and **programming languages** are key building blocks.

6. Control Theory (in AI)

- **Control theory** deals with the **behavior of dynamical systems** and how their outputs can be **modified using inputs**.
- It is widely used in **robotics** and **automation systems**.
- Helps AI systems to:
 - **Maintain stability** in changing environments,
 - Respond to **real-time inputs**,
 - Perform **feedback-based decision making**.

AI Agents

An **AI agent** is a **computer program or system** designed to:

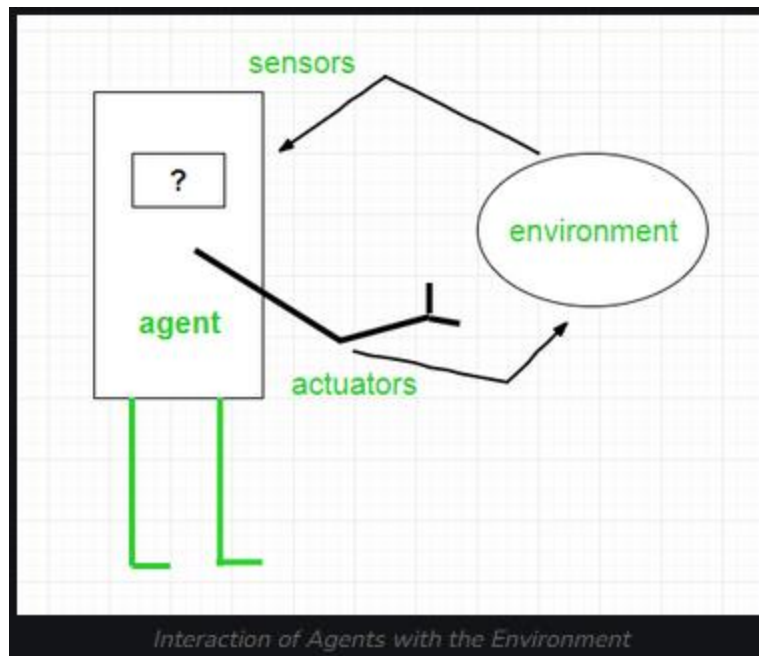
- **Perceive** the environment(Collecting Information)
- **Processing Information And Make decisions**,
- **Take actions to achieve specific goals**.
- **Learning & Improving Over Time**

OR

An **AI agent** is a software program that can interact with its surroundings, gather information, and use that information to complete tasks on its own to achieve goals set by humans.

- ❖ For instance, an AI agent on an online shopping platform can recommend products, answer customer questions, and process orders. If agent needs more information, it can ask users for additional details.

Structure of an AI Agent



An **AI agent** interacts with its environment using a clear structure made up of the following components(Sensors and Actuators)

1. Sensors

- Used to **perceive** the environment.
- They gather input data (e.g., camera, microphone, temperature sensor).

2. Agent

- The core processing unit.
- It takes input from sensors and:
 - **Processes the information**
 - **Makes decisions** based on goals and knowledge
 - Chooses appropriate actions

3. Actuators

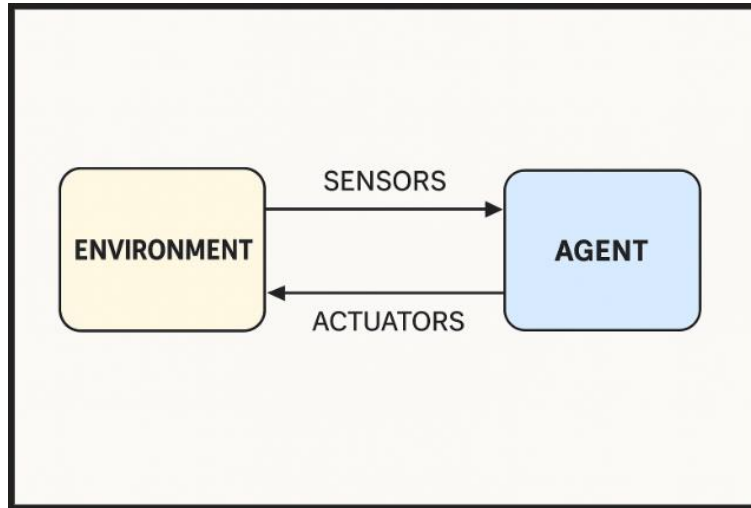
- Used to **act upon** the environment.
- Executes the decisions made by the agent (e.g., motors, speakers, display).

4. Environment

- The **external world** with which the agent interacts.
- It continuously **receives actions** and **provides feedback**.

Example:

- A smart phone assistant:
 - Uses **mic** as sensor,
 - **AI algorithm** as agent,
 - **Speakers/display** as actuators.



➤ **Agent Architecture:**

- ✚ This refers to the hardware or system that run on AI agent
- ✚ It includes Sensor (to collect information) and Actuator (to perform action)
- ✚ **EX :** smartphone is an architecture that runs on AI assistance like siri and google assistant

➤ **Agent Function :**

- ✚ Define how agent makes decision based on inputs from sensor
- ✚ **Example :** chat board remember past messages to give a better response

➤ **Agent Program:**

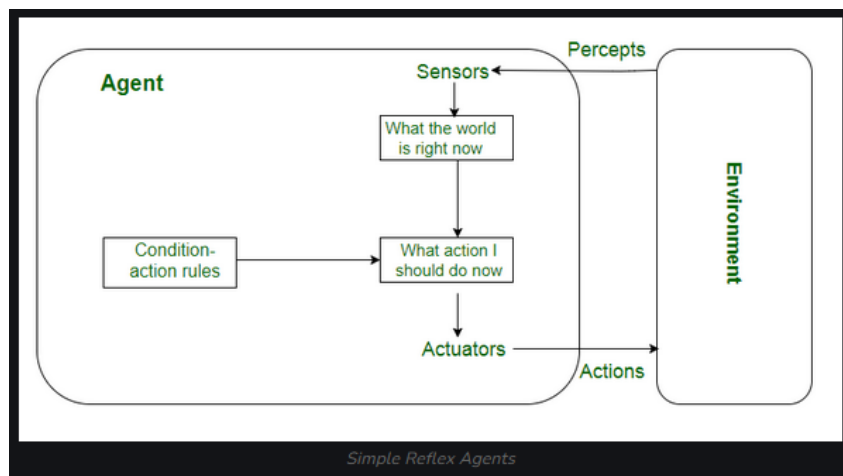
- ✚ This is a software or code make the agent function properly

Types of Agents

1. Simple Reflex Agent:

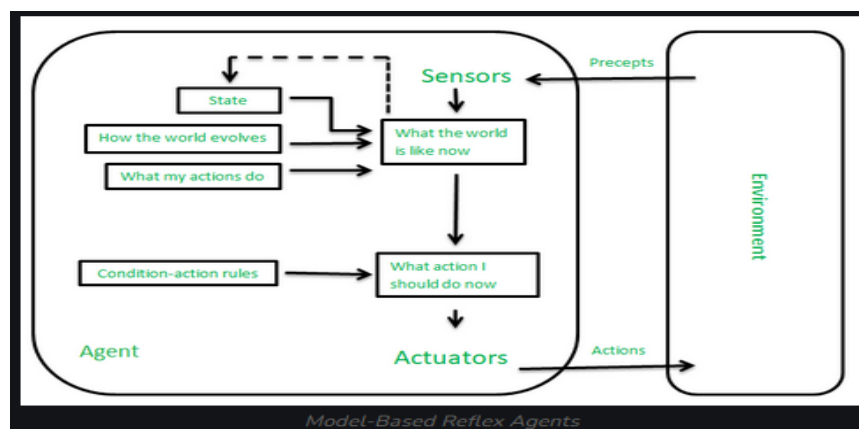
- ✚ This Agent reacts on current sensor input
- ✚ They do not remember past experience or learn from history
- ✚ They follow predefined conditions-actions rules (if the condition meets perform the action)
- ✚ **EX :**

- AI powered AC turn on cooling when temperature is high
- Motion sensor light turn on when someone walk



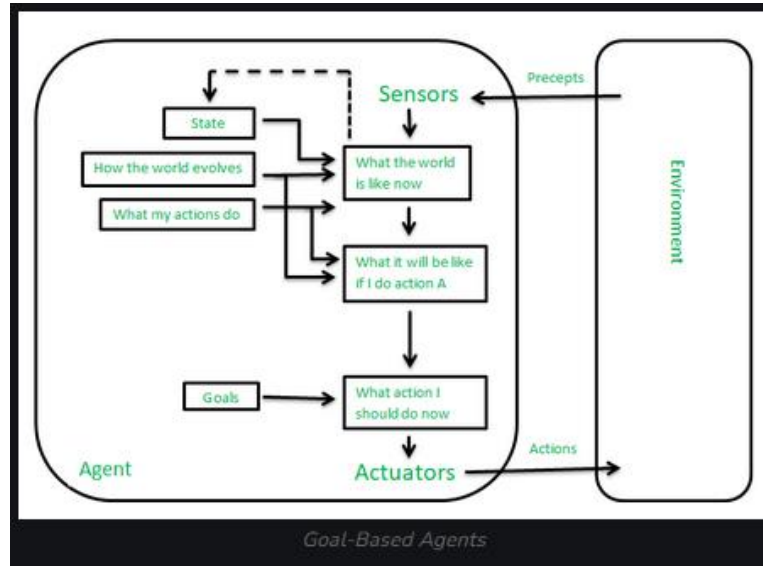
2. Model-Based Reflex Agent:

- ✚ Unlike simple reflex agent, these maintain an internal model
- ✚ They consider past experience to make better decision
- ✚ **Example :** A self driving car remember the location of even if they are out of sight



3. Goal-Based Agent

- ✚ Does agent act to achieve a specific goal instead of just reacting a condition
- ✚ They evaluate different actions and choose the best one
- ✚ **Example** : GPS navigation system



4. Utility-Based Agents

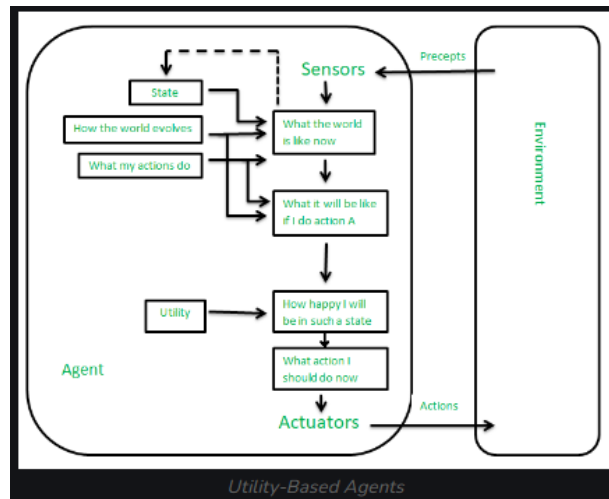
- ✚ **Chooses the best option** – Picks the action that gives the best result.
- ✚ **Thinks before acting** – Looks at different choices and decides wisely.
- ✚ **Wants to be happy** – Tries to reach a state that makes it most satisfied.
- ✚ **Smarter than simple agents** – Doesn't just react; it plans and thinks.
- ✚ **Good for tough problems** – Works well when decisions are not easy.
- ✚ **Always wants the best** – Aims to get the most benefit or value
- ✚ **Ex : Self-Driving Car** 🚗

- The car needs to decide **which route to take**.
- It checks:
 - Traffic on different roads
 - Time taken
 - Road conditions
 - Fuel usage

It **chooses the route** that is **fast, safe, and uses less fuel** — this gives the **highest utility** (best result).

So, the self-driving car **thinks like a utility-based agent**:

It **picks the smartest option** to reach the destination **happily and efficiently**.



5. Learning Agent

- ✚ **Learns from experience** (like a student).
- ✚ **Improves over time** by trying and learning from results.
- ✚ Starts with **basic knowledge**, then **gets smarter** as it learns.

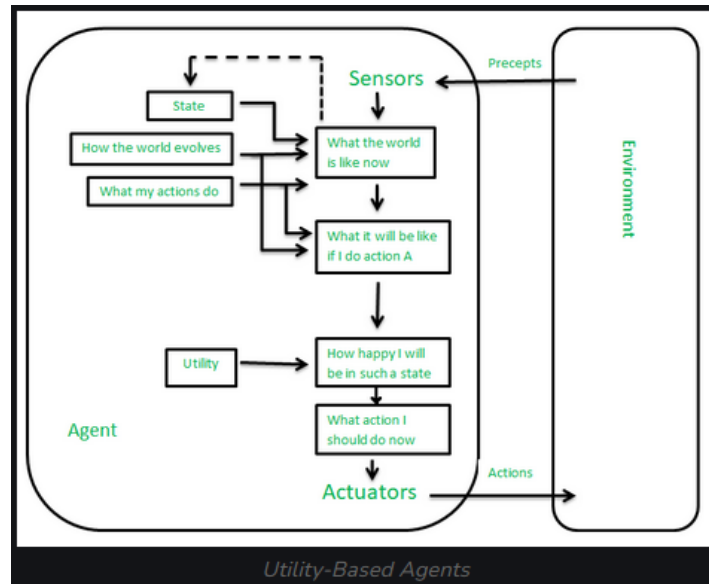
Main Parts of a Learning Agent:

1. **Learning Element**
 - Learns from experience
 - Improves how the agent works
2. **Critic**
 - Watches how well the agent is doing
 - Gives **feedback** (good or bad)
3. **Performance Element**
 - **Makes decisions** and takes actions
 - Does the actual work using current knowledge
4. **Problem Generator**
 - Suggests **new things to try**
 - Helps the agent **learn more** by exploring

Real-Life Example:

Voice Assistants (like Alexa, Siri):

learn your voice, preferences, and habits over time to improve responses and suggestions.



Environment

The environment in AI refers to the external world with which an AI agent interacts

Or

The **environment** is the **outside world** the agent operates in.

Components of an AI Agent [PESA]:

Component	Role
Performance (P)	The goal/score the agent is trying to maximize
Environment (E)	The world in which the agent operates
Sensors (S)	Used to observe the environment (input devices)
Actuators (A)	Used to take actions in the environment (output devices)

Examples

Component	Chess AI Agent	Self-Driving Car AI Agent
Environment	Chess board and opponent moves	Roads, traffic lights, vehicles, pedestrians, signs
Agent	AI playing chess	AI system inside the car controlling decisions
Sensors	Input from board state (piece positions)	Cameras, LIDAR, radar, GPS, speedometer
Actuators	Move generator (makes a move on the board)	Steering, brake, accelerator, indicators
Performance	Win the game (maximize chances of victory)	Safely reach destination in minimum time, avoid accidents

Concept of Rationality in AI

What is Rationality?

- **Rationality** means making the **best possible decision** based on:
 - Available **knowledge**
 - The agent's **goals**
 - **Perceived environment**
 - **Possible actions** and **outcomes**

A **rational AI agent** always tries to **maximize its chances of success** in achieving its goal.

How it work

- **Sees the environment** using sensors.
- **Thinks** about possible actions.
- **Chooses the best action** to reach the goal.
- **Acts** using actuators.
- (If it's a learning agent) **Learns** from results to do better next time

Examples of Rationality in AI

1. Chess AI

- **Goal:** Win the game.
- **Rational Action:** Selects the **best move** after evaluating all possible future moves.
- Uses strategy to **maximize winning chances**.

2. Self-Driving Car

- **Goal:** Reach destination safely.
- **Rational Action:** Slows down near pedestrians, stops at red lights, avoids collisions.
- Follows traffic rules and adjusts to road conditions.

Problem Solving Agent in AI

A problem-solving agent is a type of goal-based agent in which precepts are used to formulate a goal to achieve.(that find the best sequence of action to achieve specific goal)

How It Work

- i. Define the problem
- ii. Search for solution
- iii. Choose the best path
- iv. Execute the solution

Example : Navigation System AND Chess

Problem Formulation in AI

- ✚ Problem formulation is the process of defining the problem.
- ✚ To achieve a goal, a goal-based agent needs to identify a sequence of actions and states that lead to a goal state (if possible).

How It Work

- The **initial state** (where the problem starts)
- The **goal state** (what you want to achieve)
- The **actions or operators** (possible moves or steps to take)
- The **state space** (all possible states or configurations)
- The **path cost** or evaluation criteria (how to measure the cost or quality of solutions)

Example : Puzzle Problem

1. Initial State

- **What it is:** The starting arrangement or condition of the problem.
- **In the 8-Puzzle:** It is the starting layout of the 3x3 board with tiles in some scrambled order.

Example Initial State:

```
1 2 3
4 _ 5
6 7 8
```

- Here, _ (blank) means the empty space where you can slide a tile into.
 - This is where the puzzle begins.
-

2. Goal State

- **What it is:** The desired arrangement or the solution you want to reach.
- **In the 8-Puzzle:** It's the solved puzzle configuration.

Example Goal State:

```
1 2 3
4 5 6
7 8 _
```

- This is the final arrangement we want to get.
-

3. Actions (Operators)

- **What it is:** The possible moves or operations you can do to go from one state to another.
- **In the 8-Puzzle:** The moves are sliding a tile **up, down, left, or right** into the blank space.

Example Actions:

- If the blank is in the middle like this:

1	2	3
4		5
6	7	8

Possible moves are:

- Slide tile 5 to the left into the blank.
- Slide tile 4 to the right into the blank.
- Slide tile 7 up into the blank.
- Slide tile 2 down into the blank.

Only adjacent tiles to the blank can move into it.

4. State Space

- **What it is:** The set of all possible configurations you can get by applying actions repeatedly starting from the initial state.
- **In the 8-Puzzle:** All possible tile arrangements reachable by sliding tiles into the blank space.

It's like a huge map of every possible board configuration you can get during the game.

5. Path Cost

- **What it is:** A cost function that measures how expensive it is to reach a state.
- **In the 8-Puzzle:** Usually, this is the number of moves (slides) made so far.
- The goal is to find the solution with the **minimum number of moves**.